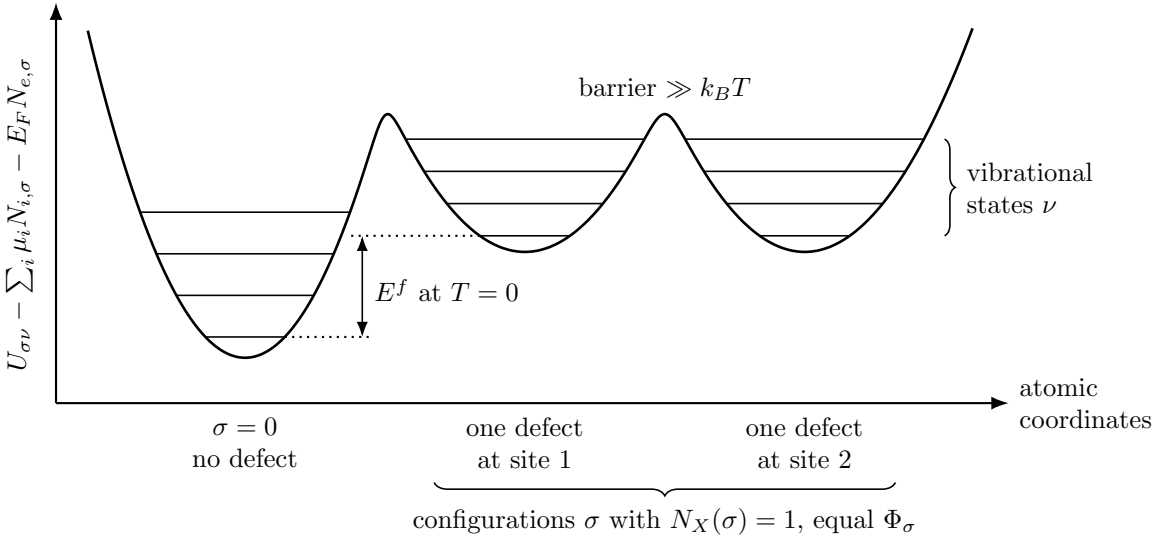

$U_{\sigma\nu} - \sum_i \mu_i N_{i,\sigma} - E_F N_{e,\sigma}$
barrier $\gg k_B T$
vibrational states ν
E^f at $T = 0$
atomic coordinates
$\sigma = 0$ no defect
one defect at site 1
one defect at site 2
configurations σ with $N_X(\sigma) = 1$, equal Φ_σ